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**A PROJECT REPORT
ON**

“PROPELLER LED DISPLAY”

Submitted in the partial fulfillment for the award of Bachelor of Engineering degree
in
COMPUTER SCIENCE & ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the Project Work entitled “**Propeller Led Display**” is the bonafide work carried out by **Chandrika H V** (4AD13CS018), **Rahul G** (4AD11CS034), **Raksha S** (4AD13CS059) and **Rakshitha S R**(4AD13CS060) in partial fulfillment for the award of degree of Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during the year 2016-2017.

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ABSTRACT

The “**Propeller LED Display**” is hardware based project. This project is a special kind of LED display. With the help of some mechanical assembly, LED count, hardware requirement, and hence overall cost is cut to very affordable price. Also, maintenance and repairing of components is so easy, that all the synchronization can be implemented through the software.

Propeller LED display is made using 40-pin 8051 series of microcontroller this project is mainly based on the pov principle and space multiplexing, this project is mechanically assembled we can display character, digit format. Made From the scrap it can be used anywhere and everywhere and gives the crystal-clear display this display consists 8 LED which are rotated in both clockwise and anti-clockwise direction to show the display.

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Chapter 1

INTRODUCTION

1. INTRODUCTION

“Propeller” is a term associated with the circular rotating object. As this project needs to rotate whole circuit assembly, there must be some prime mover attached to it, so it is termed as ‘Propeller’. This project uses bright Light Emitting Diodes (LED’s) for displaying the characters and symbols on its assembly that’s why this project is named as **“PROPELLER LED DISPLAY”**.

Conventional method of displaying images are mainly using LCD display and dot matrix where a huge number of LED’s and power processors are used to create the display. The main idea of this project is to use minimum number of LED’s and components to create a virtual display with minimum power consumption. For the purpose of displaying a set of LEDs is used. The Propeller LED display is hardware based project. This project has a special kind of LED display. Propeller LED display is made using 40-pin 8051 series of microcontroller and it is rotated in both clockwise and anti-clockwise direction to show the display. It can be used anywhere and everywhere and gives the crystal-clear display.

This is the phenomenon which is related to vision capability of human eye by which an after image is thought to persist for approximately $1/25^{\text{th}}$ of a second. So, if someone is observing the image at a rate of 25 images per second they appear to be continuous. The propeller display is a linear array of light emitting diodes, rotating at a high angular velocity to generate a circular screen. The best example of this property is the red circle we observe when we rotate the fire cracker or incense stick in circle. This project was started with the simple principle which is frequently encountered in our everyday life, which is Persistence of vision. This phenomenon makes one feel fast moving or changing objects to appear continuous. A television is a common example: in which image is re-scanned every 25 times, there by appear continuous. Further, a glowing objects if rotated in a circle at fast speed, it shows a continuous circle. By modifying this basic idea, 8 LEDs can be rotated in a circle, showing 8 concentric circles. But if these LEDs are switched at precise intervals, a steady display pattern can be shown.

Existing system do employ POV principle, but for displaying each pixel, individual LED is used. This results in a huge number of LED's even for small sized displays. By using a propeller type display, LED count can be kept to a bare minimum.

The LED's are attached to a rotating board. They turn ON and OFF at very definite and precise time intervals. All we can see are the lighted dots from the LED's making a readable display that seems to float. In this project an array of LED's, micro-controller and infrared receiver are placed on the board and are rotated by a motor at a very high rpm.

Even 8 LED's can perform a task of over 525 LED's. Applications can find their way into cost effective solutions for large public displays, information systems. It can directly replace Railway station information displays, bus stands and many more places.

Types of propeller display

1. Disc shaped
 2. Cylindrical shaped
-
1. DISC SHAPED: disc shaped display is capable of displaying analog clock.
 2. CYLINDRICAL SHAPED: cylindrical shaped display capable of displaying digits and text.

1.1 OBJECTIVE

The objectives of the project are:

- To design a display system using LEDs which rotate on a motor.
- To design a program using embedded C language.
- To display Character, date using LEDs.
- To develop software that can generate LEDs display with accurate synchronization and timing.

Chapter 2

LITERATURE SURVEY

2. LITERATURE SURVEY

➤ IEEE paper for Propeller Clock

Bhanu Pratap, Khushboo Kumari, Ankitha Raturi & Abhishek Joshi.

This project is a different view of holographic clock construct in linear array arrangement. It will be the coordination of electrical, electronics and some mechanical engineering. This “illusion” is based on inertia of human eye. The motor spins at a constant rate such that they will display the current time. The motor spinning fast enough that the human eye will perceive all of the display is on at once, and the viewer will be able to read the time constantly. If LED formed digits will periodically and frequently enough flash, they will appear solid and steady.

A micro controller is used to keep the time and blink the LED's in an appropriate pattern to show the numbers. It has to be programmed so that it will both keep time and also send the appropriate signals to the LED's to light them in the correct sequence. It looks like the digital numbers are floating in the air.

➤ Propeller LED display for images

A.N.V.K. Chaitanya, Dept of ECE.

Propeller display is a special kind of circular LED display. It is making use of POV, Persistence of Vision, which means that if something appears in the same spot consistently, at least 50-60 times per second, our brains think that it's permanently there when it is not. Conventional methods of displaying images to public are LCD display and dot matrix LED board. Propeller LED display is a device that project an image or time as if the images are floating in the air. The floating image is received because of human eye limitations. The floating images emerge by synchronizing LED's blink to form an image at particular time and rate. This propeller display is mechanically scanned and displays in digital format. Made from scrap it can be used anywhere and everywhere and the interesting fact is that its crystal clear display. Maintenance and repairing of the display is so easy. All the synchronizing can be implemented through software.

➤ **About 8051 Micro-controller, an application based introduction**

Chris Braithwaite, Fred Cowan, Hassan Parchizadeh.

A microcontroller is a computer with most of the necessary support chips on board. All computers have several things in common, namely:

1. A central processing unit (CPU) that ‘executes’ programs.
2. Some random-access memory (RAM) where it can store data that is variable.
3. Some read only memory (ROM) where program to be executed can be stored.
4. Input and output (I/O) devices that enable communication to be established with the outside world i.e. connection to devices such as keyboard, mouse, monitors and other peripherals.

There are a number of other common characteristics that define microcontrollers. If a computer matches a majority of these characteristics, then it can be classified as a microcontroller. Microcontroller may be:

1. “Embedded” inside some other device so that they can control the feature or actions of the product. Another name for a microcontroller is therefore an ‘embedded controller’.
2. Dedicated to one task and run one specific program. The program is stored in ROM and generally does not change.
3. A low power device. A battery-operated microcontroller might consume as little as 50 mill watts.

➤ **Selection and Using of DC Motor Specification**

105 Webster St. Hanover Massachusetts

The functional requirements and design parameters of a motor system can be determined early in the design process and manufacturer’s published motor data can be researched in an attempt to find suitable match.

Designers and engineers use mathematical models to optimize the time spent designing modules, mechanism by:

1. Reducing the number of possible items in selection set.

2. Reducing testing and prototyping time.
3. Providing a closer match of motor performance to the functional requirements and design parameters specified early in the design process.

➤ **Keil software**

www.MicroDigitalEd.com

Micro Vision, the popular IDE from Keil Software Projects Management, Source Code Editing, Program Debugging, and Flash Programming in a single, powerful environment. This Quick start give information necessary to use micro Vision 3 for own projects. It provides an overview of the most commonly used micro Vision 3 features including:

1. Project Management, Devices Setup, and Tool Configuration.
2. Editor facilities for Creating, Modifying, and Correcting Programs.
3. Target Debugging or CPU & Peripheral Simulation.

➤ **IR Sensors circuit diagram and working**

An IR sensor is an electronic device that emits in order to sense some aspect of surroundings. IR sensors can measure the heat of an object as well as detect the motion. These types of a sensor measures only infrared radiation, rather than emitting it that is called as passive sensors. The emitter is simply an IR LED and detector is simply an IR Photodiode which is sensitive to IR light of the same wavelength as that emitted by the IR LED. When IR light falls on the photodiode, the resistance and these output voltages, change in proportion to the magnitude of IR light received.

An infrared sensor circuit is one of the basic and popular sensor model electronic devices.

➤ **Embedded C**

Michael J. Pont

An embedded system is an application that contains one programmable computer. Typically in form of microcontroller and which is used by individuals who are, in the main, unaware that the system is computer based.

The important decisions that made at the start of any embedded project:

1. The choice of processor
2. The choice of programming language.
3. The choice of operating system.

Example of the embedded application that are constructed using the techniques that are

1. Mobile phone systems.
2. Automotive application.
3. Domestic appliances.
4. Aerospace application.
5. Medical equipment.
6. Defense system

Chapter 3

REQUIREMENT SPECIFICATION

3. REQUIREMENT SPECIFICATION

3.1 HARDWARE REQUIREMENTS:

- RAM : 4GB
- Hard disk drive : 1TB
- DC Motor : 500 RPM
- LED : 8 LEDs
- Microcontroller 8051 : P89V51RD212
- Sensor : IR sensor
- Battery : lithium ion 5v
- Oscillator : crystal oscillator
- Printed circuit board

3.2 SOFTWARE REQUIREMENTS:

- Tool: KEIL software.

The Keil 8051 development kits are a complete solution for creating software for the 8051 family of microcontroller. The development kits comprise many different tools that allow projects ranging from simple to highly complex to be developed with relative ease. The Keil C51 C compiler for the 8051 microcontroller is the most popular 8051 C compiler in the world. It provides more features than any other 8051 C compiler available today. The C51 compiler allows you to write 8051 micro controller applications in C that, once compiled, have the efficiency and speed of assembly language. Language extensions in the C51 compiler give you full access to all resources of the 8051. The C51 compiler translates C source files into relocatable object modules which contain full symbolic information for debugging with the micro vision debugger or an in-circuit emulator. In addition to the object file, the compiler generates a listing file which may optionally include symbol table and cross reference

information. We used the Keil compiler to write the program code. The code is written in C language for the word GREIT.

- Language : Embedded C
- Operating System : Windows XP/7

Chapter 4

SYSTEM ANALYSIS

4. SYSTEM ANALYSIS

4.1 PROBLEM STATEMENT

From other article and projects, there are some problems which are affected the outcome of the image display. The main problem is concerning the motor. A VCR motor has to be used as a motor in order to prevent noise. The circuit also has to use phototransistor to generate precise index pulse. In order to display the date and clock, the circuit must have memory capacitor to keep the clock ticking even when the power is not supplied to the circuit. The memory capacitor is expensive plus, it's bigger and heavy. To replace the memory capacitor, a small, cheap and liter 3V lithium battery is used as a replacement. A voltage regulator also has to be used in order to allow rotation speed change without affecting the circuit and displayed images.

4.2 EXISTING SYSTEM

Existing systems do employ POV principle, but for displaying each pixel, individual LED is used. This results in a huge number of LEDs even for small size displays. By using a propeller type display, LED count can be kept to a bare minimum. Even 7 LEDs can perform a task of over 525LEDs.

4.3 PROPOSED SYSTEM

We are using single columns of LEDs and rotating it in a clockwise so first column pattern must be displayed and some delay must be maintained to control the LEDs ON time of a column.

If delay is not maintained we cannot see the LED glow.

Problem with Existing system is that, we are using each pixel for displaying each pixel individual LED's. This results in huge number of LED even for small display, which in turn increases in cost.

This problem can be overcome by using Propeller type Display. In this we are using single column of LED's and rotating it in clockwise direction. So first column pattern must be displayed and some delay must be maintained to control the LED's on time of a column. If the delay is not maintained we can't see the LED glow.

The project which is a special kind of circular LED display. With help of some mechanical assembly, LED count and hardware requirement all the synchronizing can be implemented through software. First of its kind, made using 20pin 8051 series micro-controller, this project use the principal of space multiplexing. It can be used anywhere and everywhere and display is its crystal clear display.

The display consists of 8 bright LED's which are rotated to show the display which is fixed in another side of arm. When DC Motor is rotating, the character will appear. The synchronization of DC Motor speed and LED blink cause the character visible to human eyes.

Advantages of Propeller LED Display:

- Requires less LED
- Cost decreases
- Space Multiplexing
- Crystal Clear display

4.4 COMPONENTS

1. DC-motor
2. Battery
3. LEDs
4. Microcontroller
5. IR Sensors
6. Oscillators

4.4.1 DC Motor:

DC motors are non-polarized which means that it can reverse voltage so the motor will rotate in two directions, forward and backward. DC motor rated from about 6V-12V. The large one are, often 24V, for this project it is necessary to use 6V-12V range motor. Specification of most DC motors shows high rpm (revolutions per minute) and low torque.



Fig 4.4.1 DC Motor

4.4.2 LED:

A light emitting diode is a semiconductor device that emits a visible light when an electric current is passed through it. It is monochromatic, occurring at single wavelength.

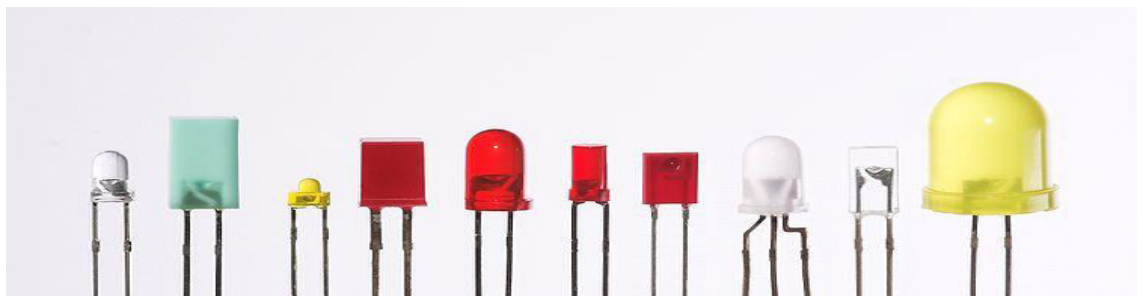


Fig 4.4.2 Light Emitting Diode

Benefits of LED:

1. Low power requirement: Most types can be operated with battery power supply.
2. High efficiency: a minimal heat production.
3. Long life: when properly installed on LED can function for decades.

Applications:

1. Indicator lights
2. LCD panel backlighting
3. Fibre optic data transmission
4. Remote control

4.4.3 INTERRUPT MODULE:

Interrupter module is our sensor module, consisting of the IR interrupt sensor. This sensor was selected from a variety of other alternatives, because of its small size, precise interrupt sensing, and study casing. One great advantage of using this module is, interfacing it with the microcontroller is just a matter of two resistors and a general purpose transistor. MOC7811 is the sensing part of the interrupter module, while rest of the circuitry works as signal conditioning circuit, wires emerge out from the module, respectively Vcc, Signal and Ground. Output of the module is LOW, if interrupt occurs, otherwise it remains HIGH. It consists of IR LED and Photodiode mounted facing each other enclosed in plastic body. When light emitted by the IR LED is blocked because of some completely opaque object, logic level of the photodiode changes. This change in the logic level can be sensed by the microcontroller or by discrete hardware.

IR sensor basically a device which consists of a pair of IR LED and photodiode which are collectively called a photo-coupler. The IR LED emits IR radiation reception, by the photodiode detects the output of the sensor.

This sensor module work on the principle of reflection of infrared rays, a continuous beam of IR rays is emitted by the IR LED whenever a reflecting surface come in front of receiver, these rays are reflected back and captured.

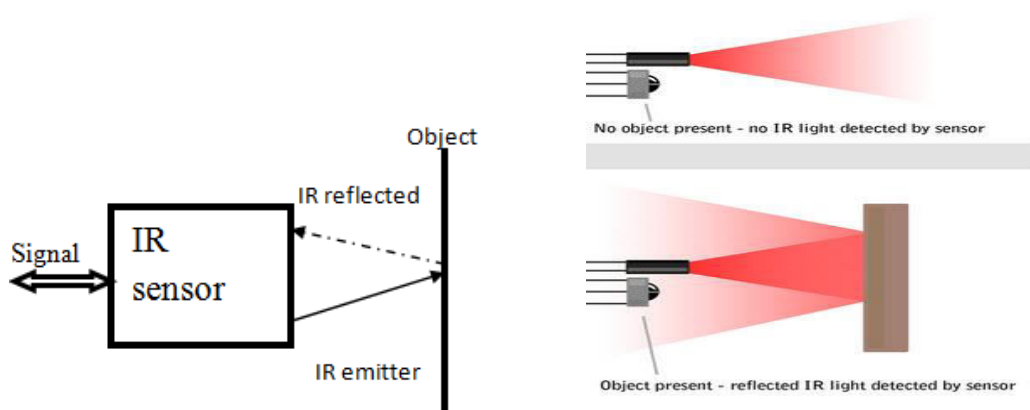


Fig 4.4.3 Interrupts Module

4.4.4 MICRO-CONTROLLER (8051)

History: In 1981, Intel Corporation introduced an 8-bit microcontroller called the 8051. This microcontroller had 128 bytes of RAM, 4K bytes of on-chip ROM, two timers, one serial port, and four ports (each 8-bits wide) all on a single chip. At the time it was also referred to as a “system on a chip.” The 8051 is an 8-bit processor, meaning that the CPU can work on only 8 bits of data at a time. Data larger than 8 bits has to be broken into 8-bit pieces to be processed by the CPU. The 8051 has a total of four I/O ports, each 8 bits wide. Although the 8051 can have a maximum of 64K bytes of on-chip ROM, many manufacturers have put only 4K bytes on the chip.

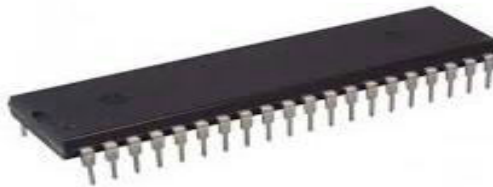


Fig 4.4.4 8051 Microcontroller

4.5 ARCHITECTURE OF 8051

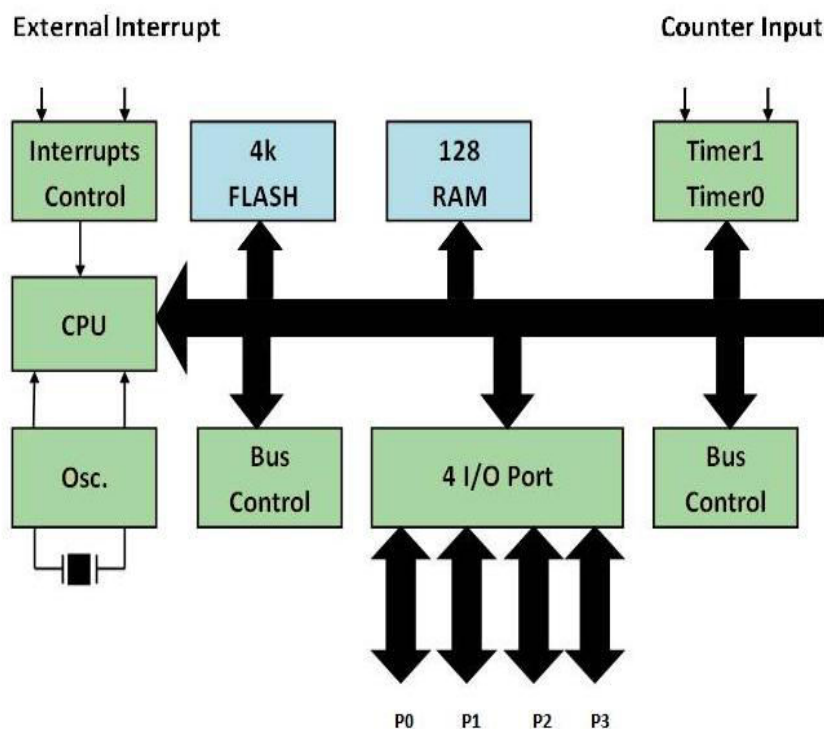


Fig 4.5 Architecture of 8051

4.5.1 Interrupt control:

The Microcontroller 8051 can be configured in such a way that it temporarily terminates or pause the main program at the occurrence of interrupts.

When subroutine is completed the execution of the main program starts.

The five interrupt sources are there in 8051 microcontroller:

1. INT0
2. TF0
3. INT1
4. TF1
5. R1/T1

Out of these INT0 and INT1 are external interrupts.

When all these interrupts are activated set the corresponding flags except for serial interrupt,

1. The interrupt flags are cleared when the processor branches to the interrupt service routine (ISP).
2. The external interrupt flags are cleared when the processor branches to the interrupt is a negative edge triggered where as the timers and serial port interrupts two of them are external interrupts, two of them are timer interrupt and one serial port interrupt terminal in general.

4.5.2 Memory

Microcontroller requires a program, which is a collection of instruction. This program tells microcontroller to do specific task, these can be saved and read by microcontroller to perform specific operation.

1. The memory which is used to store the program of the microcontroller is known as code memory or program memory also ROM memory.
2. The data memory of 8051 is used to store data temporarily for operation is known as RAM memory.
3. 8051 microcontroller has 4k of code memory, that has 4kb ROM and also 128 bytes of data memory of RAM.

4.5.3 CPU

It monitors and controls all operations that are performed in the microcontroller. User have no control over the work of CPU. It reads program written in ROM memory and execute them and do the expected task.

4.5.4 Bus

Bus is a collection of wires which work as a communication channel or medium for transfer of data. These buses consist of 8,16 or more wires thus these can carry 8 bits, 16 bits simultaneously.

Buses are of two types:

1. Address bus
2. Data bus

Address bus: 16 bit address bus it used to address memory location. It is used to transfer the address from CPU to memory.

Data bus: It has 8 bit data bus it is used to carry data.

4.5.5 Oscillator

Microcontroller is a digital circuit device, therefore it requires clock for its operation. Microcontroller 8051 has an on-chip oscillator which works as a clock source for CPU. As output pulses of oscillator are stable therefore it enables synchronized work of all parts of 8051 microcontroller.

4.5.6 I/O ports

Microcontroller is used in embedded system to control the operation of machines. Therefore to connect it to other machines, device or peripherals we require is input output interfacing ports in microcontroller.

Microcontroller 8051 has 4 input output ports to connect it to other peripherals.

4.5.7 Timers/Counters

8051 has two 16 bit timers and counters. The counters are divided into 8 bit registers. Timers are used for measurement of intervals to determine pulse width etc.

4.6 PHILIPS FAMILY

Another major producer of the 8051 family is Philips Corporation. Indeed, they have one of the largest selections of 8051 microcontrollers. Many of their products include features such as A-to-D converters, D-to-A converters, extended I/O, and both OTP and flash.

P89V51RD212

(ISP) and In-Application Programming (IAP) capability

- Boot ROM contains low level Flash programming routines for

Downloading via the UART

- Can be programmed by the end-user application (IAP)
- 6 clocks per machine cycle operation (standard)
- 12 clocks per machine cycle operation (optional)
- Speed up to 20 MHz with 6 clock cycles per machine cycle (40 MHz equivalent performance); up to 33 MHz with 12 clocks per machine cycle
- Fully static operation
- RAM expandable externally to 64 Kb
- 4 level priority interrupt
- 8 interrupt sources
- Four 8-bit I/O ports
- Full-duplex enhanced UART
 - Framing error detection
 - Automatic address recognition
- Power control modes
 - Clock can be stopped and resumed
 - Idle mode
 - Power down mode
- Programmable clock out
- Second DPTR register
- Asynchronous port reset
- Low EMI (inhibit ALE)

4.6.1 Pin diagram

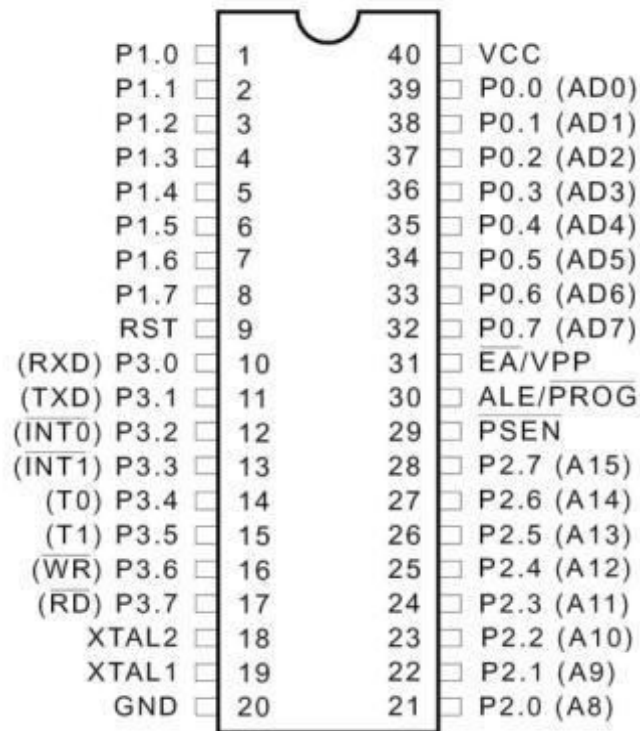


Fig 4.6.1(a) Pin Diagram

VCC : Supply voltage.

GND : Ground

PORT 1

Each of these pins can be configured as input or output pins.

RST

It is a active high signal, when a pulse(square wave) is applied to this pin microcontroller will terminate all its activities and reset.

PORT 3

Each of these pins can be configured as input or output pins.

RXD AND TXD

8051 are serial data communication circuit that use SBUF register to hold the data and SCON

to control the data communication.

The data is transmitted out of 801 through the TXD line.

The data is received out of 801 through the TXD line.

INT0 AND INT 1

Interrupt 0 and interrupt 1 are two input pin that are triggered by external circuit.

T0 AND T1

The 8051 has two 16-bit timers/counter.

T0 – timer 0 register, T1 – timer 1 register

They can be used either as timers to generate a time delay or as counters to count event happening outside the micro controller. Each 16-bit registers can be accessed as two separate 8-bit registers.

$\overline{\text{RD}}$ AND $\overline{\text{WR}}$

These are active low pin

When $\overline{\text{RD}}=0$, microcontroller reads the data from external RAM

When $\overline{\text{WR}}=0$, microcontroller writes the data from external RAM.

PORT 2

If external memory is not used, these pins can be used as input or output.

If external memory is used then the higher address that is A8-A15 will appear on this port.

$\overline{\text{PSEN}}$

If memory access is from the byte of a program code in the ROM, then $\overline{\text{PSEN}}$ signal goes low and the data byte from the ROM is placed on the data bus.

ALE

When ALE =1, port 0 is providing lower order address (A0-A7)

When ALE =0, port 0 is used as a data line.

OSCILLATOR CHARACTERISTICS:

The 8051 has an on-chip oscillator but require an external clock to run it. A quartz crystal oscillator is connected to input XTAL1 and XTAL2 with two capacitors having value 30pf.

If an external frequency (from AF0) have to be applied, then it must be applied between XTAL1 and GND. XTAL2 must be left open.

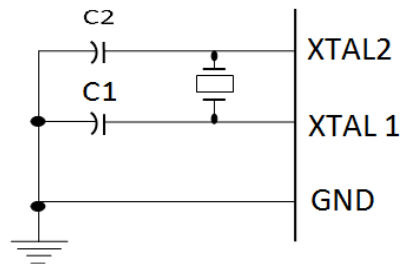


Fig 4.6.1(b) oscillator connections

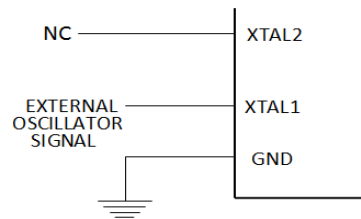


Fig 4.6.1(c) external clock drive configuration

$\overline{EA}$

This pin is used whenever external memory is used.

When $\overline{EA}$ is connected to V_{cc} that is $\overline{EA}=1$, code is stored in internal RAM

When $\overline{EA}$ is connected to GND that is $\overline{EA}=0$, code is stored in external RAM.

PORT 0

If the external memory is not used then this pin is used as input or output.

If external memory is used then lower address that is A0-A7 will appear on this port

Chapter 5

SYSTEM DESIGN

5. SYSTEM DESIGN

5.1 BASIC PRINCIPLE BEHIND THIS PROJECT:

5.1.1 POV (Persistence Of Vision):

This is the phenomenon which is related to vision capability of human eye by which an after-image is thought to persist for approximately $1/25$ th of a second. So, if someone is observing the images at a rate of 25 images per second, then they appear to be continuous. The best example of this property is the red circle we observe when we rotate the firecracker or incense stick in circle.

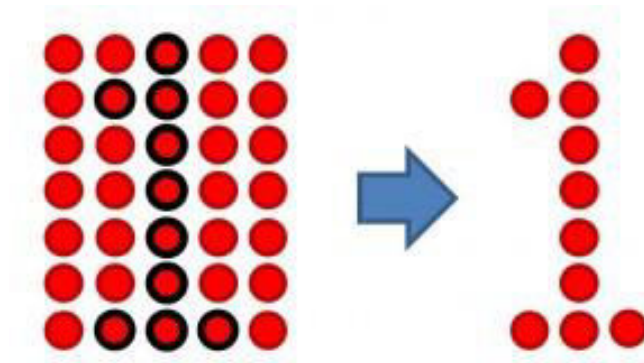


Fig 5.1 Persistence Of Vision

5.1.2 LED_DISPLAY:

This project requires bright light emitting diodes for displaying the characters on its assembly. That's why this project is named as 'PROPELLER LED DISPLAY'.

5.2 BLOCK DIAGRAM

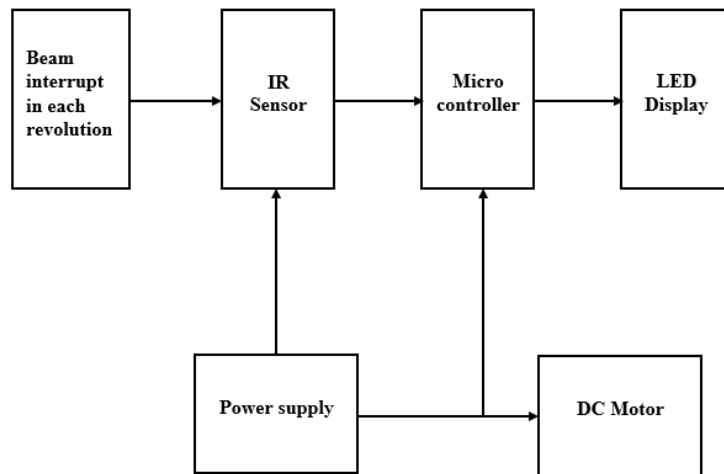


Fig 5.2 Block Diagram

This section emphasize on detailed overview of each of the block shown above.

The main important part of the propeller display is microcontroller, we are using microcontroller P89V51RD2. We can also use the other like ATmega32A, and 8051. The IR LED, which is stationary, is fixed on the base of rotating assembly. When the IR rays send by the IR LED falls on the photo diode, which is placed on the rotating PCB, it will generates a low pulse on the interrupt pin of microcontroller which results in the generation of desired pattern.

The anodes of LEDs are joined together and connected to 5v supply. Logic 0 level at the output of port0 will result in illumination of LED.

5.3 MECHANICAL ASSEMBLY

Mechanical assembly plays a vital role in proper functioning of this project. The display is scanned each time, by rotating the whole assembly in a circular path. The basic idea developed by implementing and modifying different ways. The display is displays the character by rotating the whole assembly in a circular path. Following diagram shows the most reliable way path.

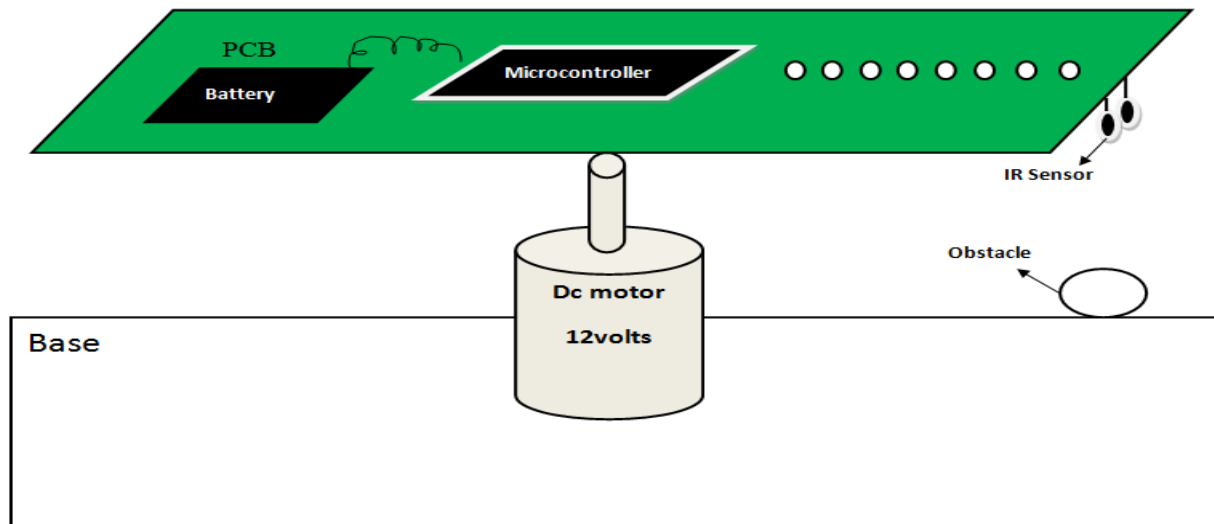


Fig 5.3 Block arrangement of components

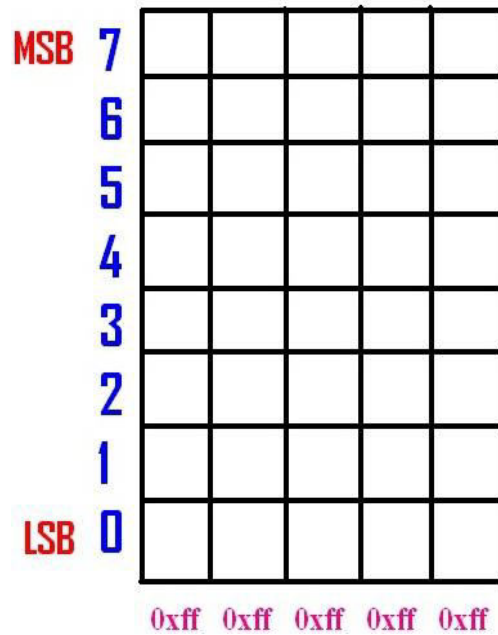
Here, one major challenge was how to bring +5V to spin circuit. We tried the same by adopting two-three different method, but finally concluded on by using 5V battery mounted on PCB.

Power supply

We are using a 12V motor dc motor to rotate the assembly to provide 9v to the DCmotor we have used a voltage adapter, which has an in built rectification circuit, that converts 230 V AC mains to 12V DC. The 12V DC is given to an arbitrary PCB then with connectors the supply is given to motor. Microcontroller we are using, P89V51RD2, operates at 5V DC supply. The DC supply is provided by using 5V battery.

5.4 HOW TO DISPLAY A CHARACTER

For displaying each letter make a standard notation. We are following 5*7 notations.



Example to display the character “A”

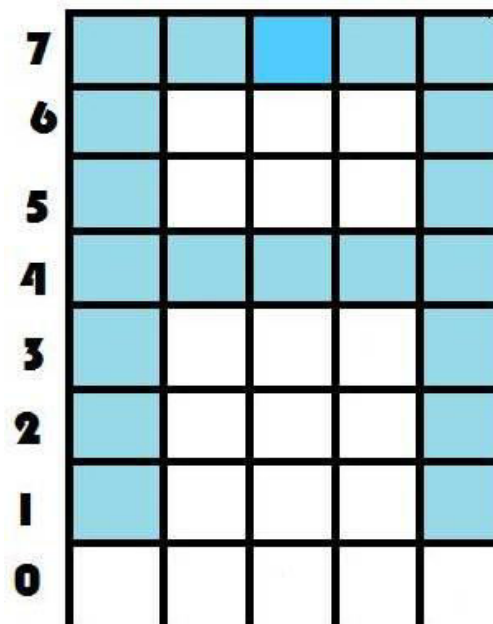


Fig 5.4(a) Character A

Input to the microcontroller given in the form of hexadecimal, which is taken by dividing the 8bit (8 LEDS) into 4bit of MSB and 4 bit of LSB shown below.

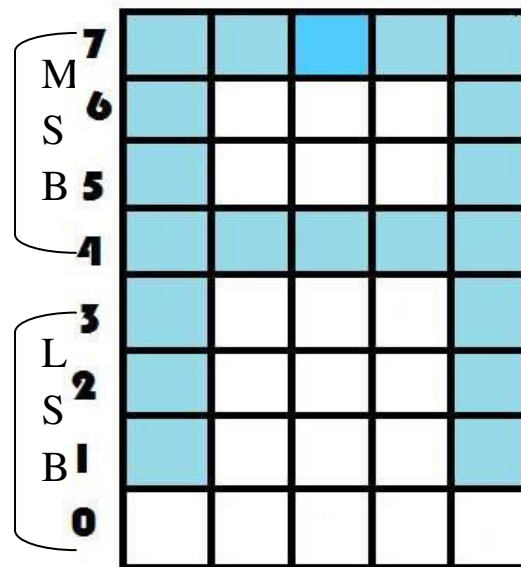


Fig 5.4(b) Representation using LSB & MSB

Blue color indicates LED in ON mode.

White color indicates LED in OFF mode.

Proper delay must be maintained to observe the characters, without the delay it is not possible to observe the clear display.

If motor is rotating in clock wise direction then the corresponding code will be as follows.

LEDs are connected to PORT0 of AT89C51rd so the logic will be

```
P0=0x7f; delay ( ); //define this function as per your motor
P0=0x88; delay ( );
P0=0x88; delay ( );
P0=0x88; delay ( );
P0=0x7f; delay ( );
```

If it is anti clock wise then same code in the reverse order in this example we are using the letter "A" which is symmetrical

```
led = 0x00;delay();column gap between 2 letters
```

Example for displaying the string ATME COLLEGE in matrix form

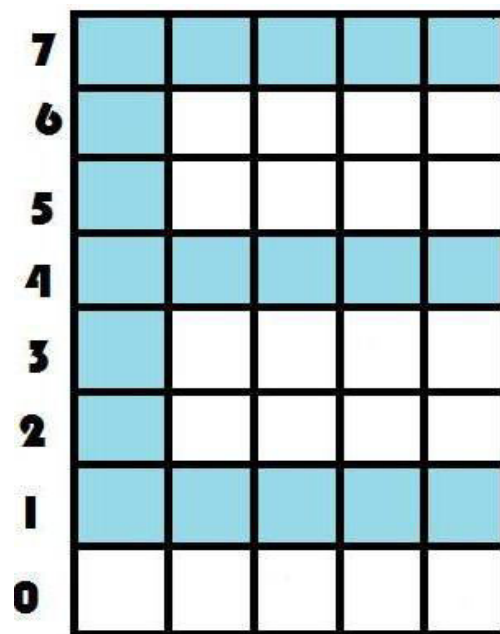
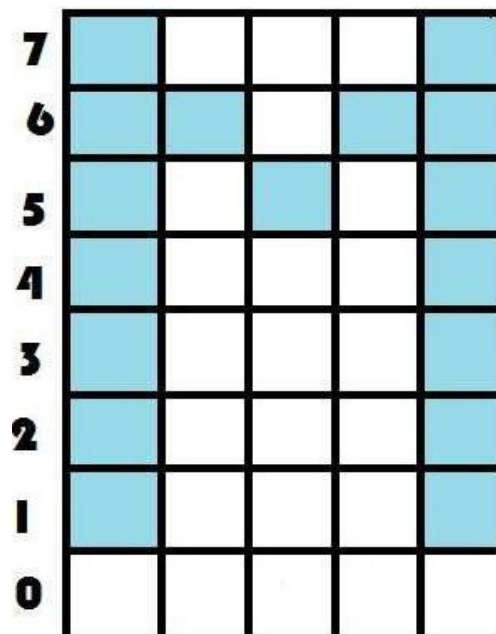
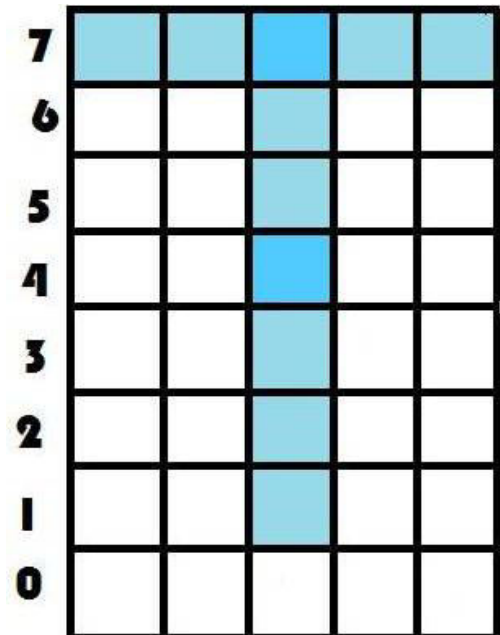
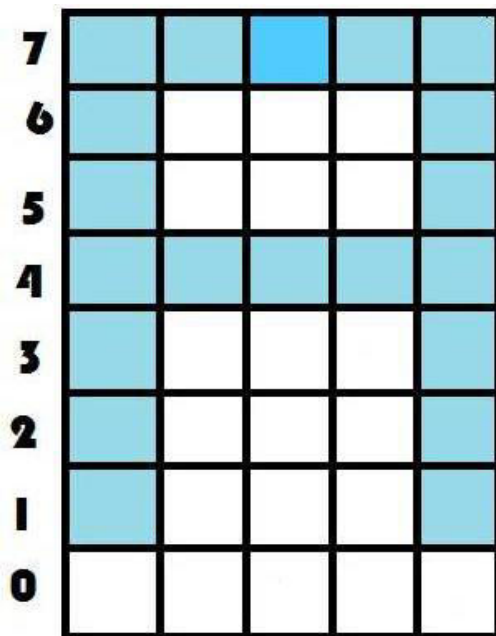


Fig 5.4(c) Characters A T M & E

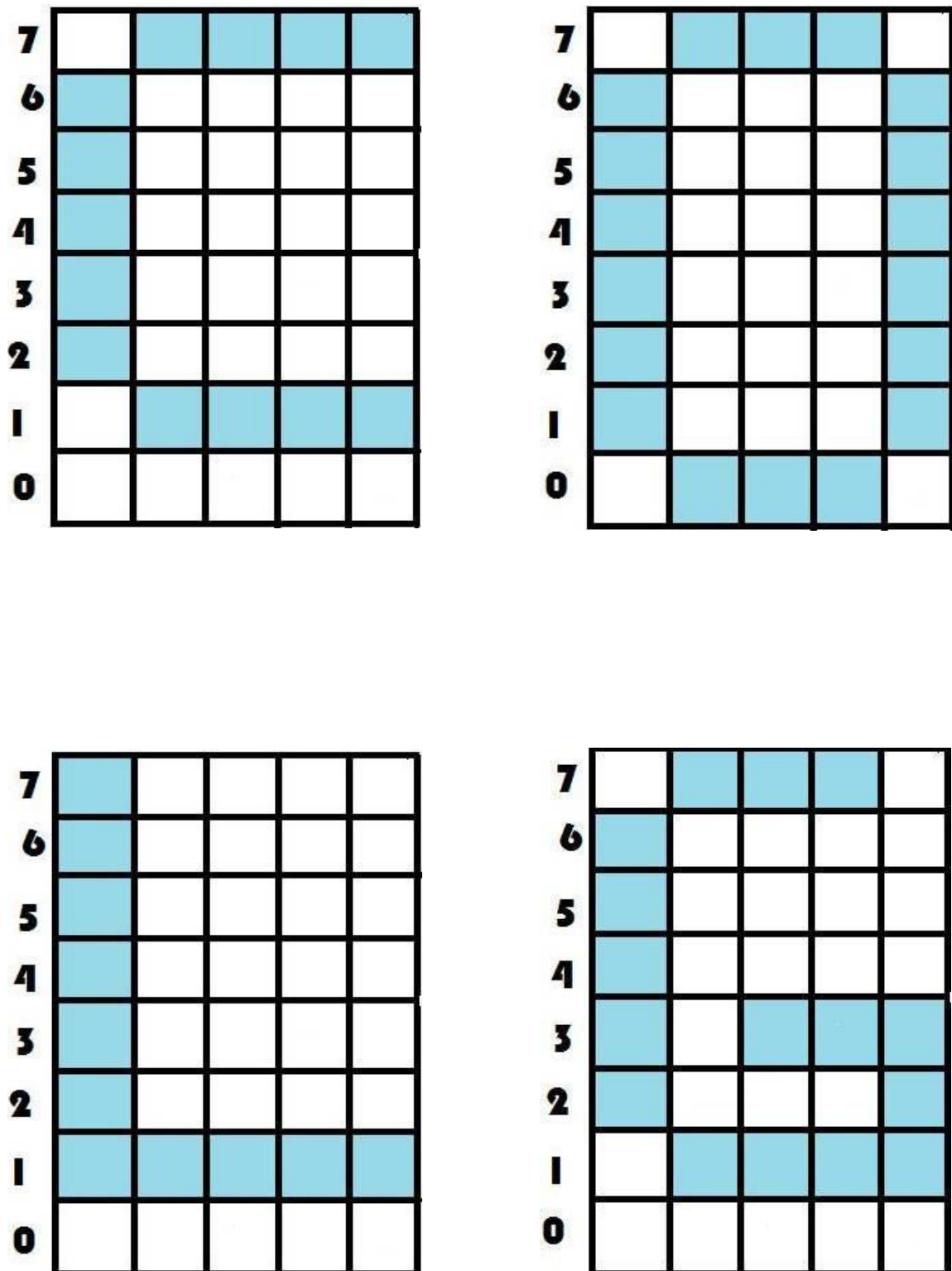


Fig 5.4(d) characters C O L & G

Chapter 6

IMPLEMENTATION

6. IMPLEMENTATION

6.1 INTRODUCTION TO KEIL MICRO VISION (IDE)

Keil an ARM Company makes C compilers, macro assemblers, real-time kernels, debuggers, simulators, integrated environments, evaluation boards, and emulators for ARM7/ARM9/Cortex-M3, XC16x/C16x/ST10, 251, and 8051 MCU families.

Keil development tools for the 8051 Microcontroller Architecture support every level of software developer from the professional applications engineer to the student just learning about embedded software development. When starting a new project, simply select the microcontroller you use from the Device Database and the μ Vision IDE sets all compiler, assembler, linker, and memory options for you.

Keil is a cross compiler. So first we have to understand the concept of compilers and cross compilers. After then we shall learn how to work with keil.

From the C program to the machine language

The C source code is very high level language, meaning that it is far from being at the base level of the machine language that can be executed by a processor. This machine language is basically just zero's and one's and is written in Hexadecimal format, that why they are called HEX files.

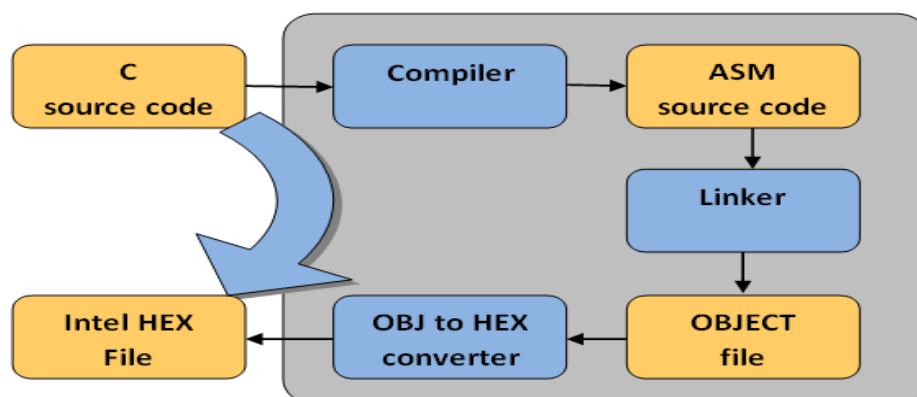


Fig 6.1 Conversion of C Program to HEX file

There are several types of HEX files; we are going to produce machine code in the INTEL HEX-80 format, since this is the output of the KEIL IDE that we are going to use. Fig shows that to convert a C program to machine language, it takes several steps depending on the tool you are using, however, the main idea is to produce a HEX file at the end. This HEX file will be then used by the ‘burner’ to write every byte of data at the appropriate place in the EEPROM of the P89V51RD2.

6.2 USING THE KEIL ENVIRONMENT

KEIL μ Vision is the name of software dedicated to the development and testing of a family of microcontrollers based on 8051 technology, like the P89V51RD2.

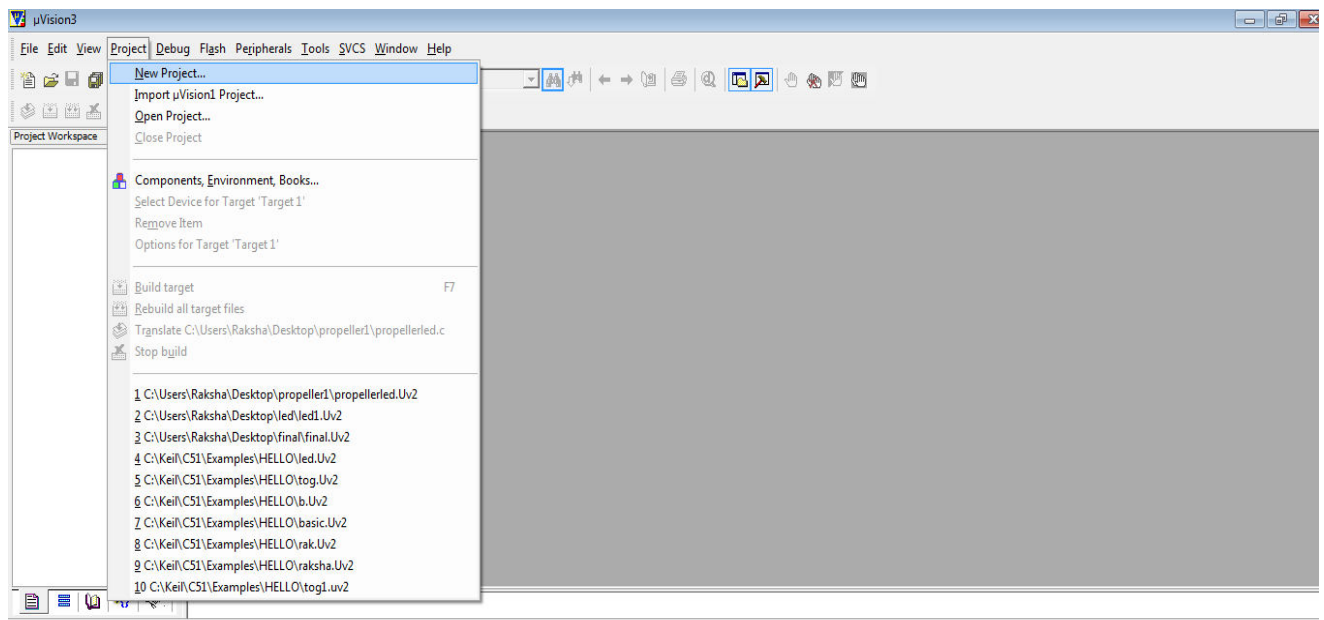


Fig 6.2 Selecting New Project

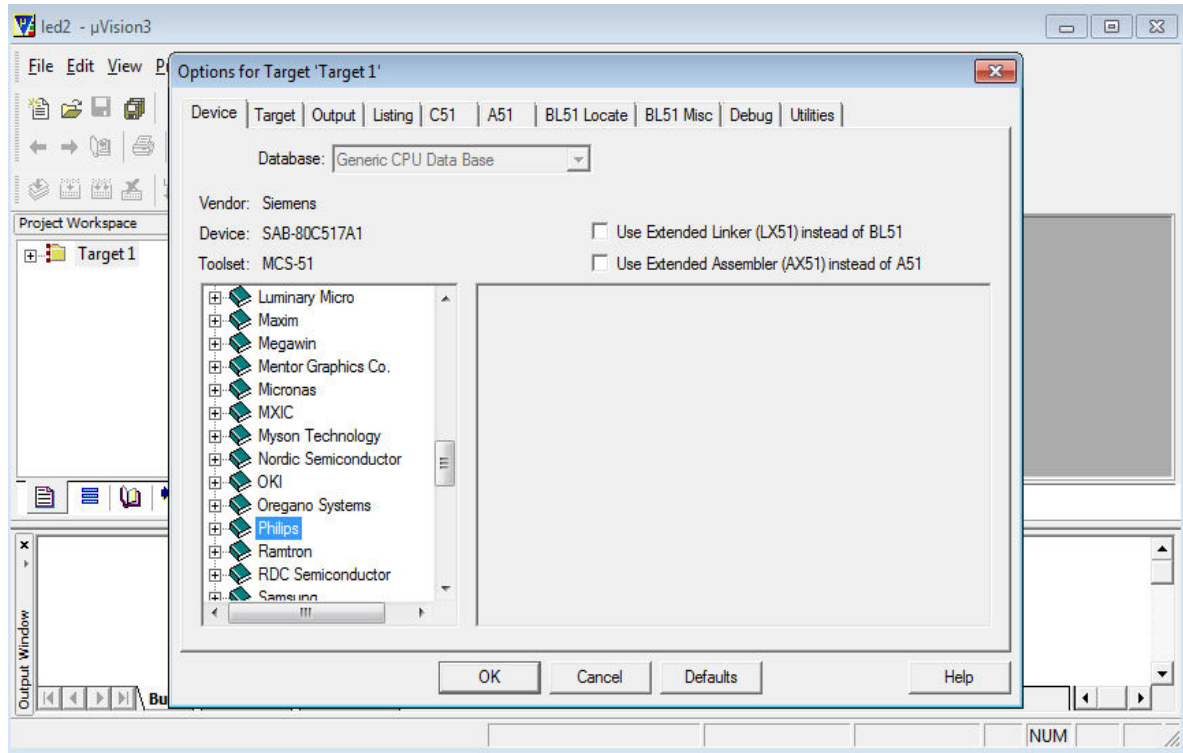


Fig 6.2.1 Selecting Device

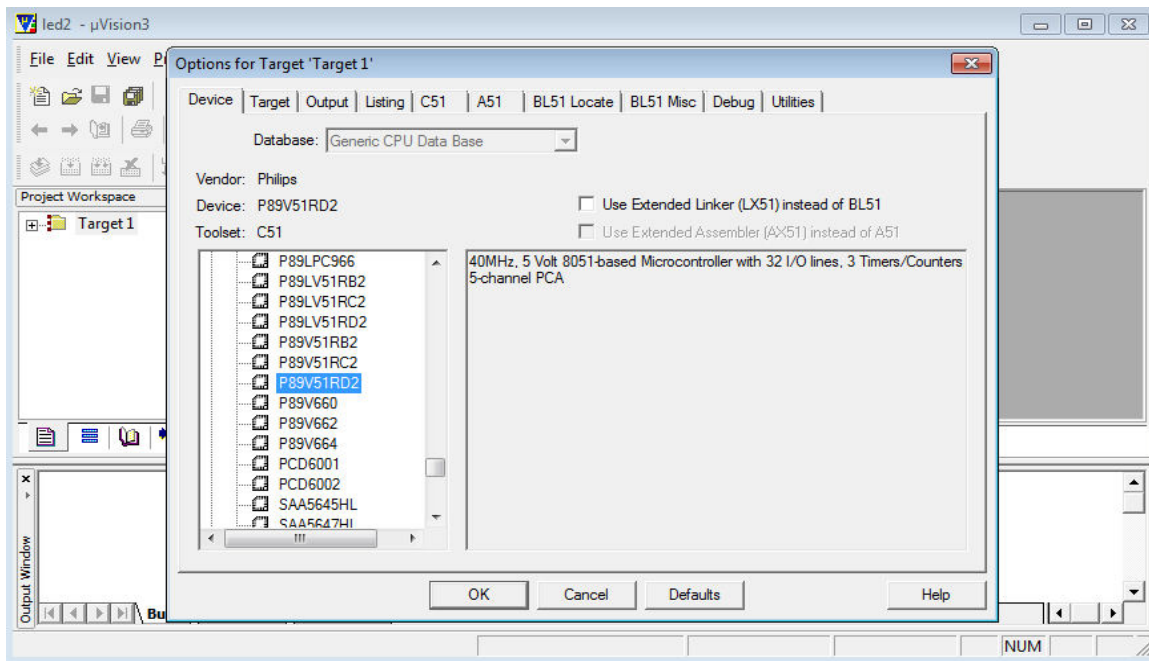


Fig 6.2.2 Selecting IC

6.3 IMPLEMENTATION

We are using 8051 microcontroller instead of Arduino board which is mounted on PCB board which in turn consists of battery, LEDs.

We are using 11.0592MHz crystal and a 7805 for power supply, LEDs are connected to the PORT0 of the MCU. 8 LED's are used for displaying letters. LEDs are connected to the microcontroller in two ways

1. ACTIVE HIGH (logic '1' on MCU pin will make the LED glow)
2. ACTIVE LOW (logic '0' on MCU pin will make the LED glow)

It is better to connect the LEDs cathode to MCU pin and anode to Vcc through 220ohm resister for current limiting.

DELAY CALCULATIONS

Calculation of proper delay is very important in making this project, because speed of any two motors is not equal. Calculate delay as per your motor. Motor speed should not be less than 1000.

motor speed===500 RPM

time for one rotation===1.2milli seconds

radius =12cm

perimeter= $2 \times 3.14 \times 12 = 75.36 \sim 75$

width of led column=0.5cm (this indicated the duration of led glow in terms of length of display)

total num of columns(leds)= $75/0.5=150$

150 leds=1.2 milli seconds

One led(column)time=50micro second

columns for each letter=5

time for a letter= $5 \times 50 = 250$ micro seconds

length for letter= $5 \times 0.5 = 2.5$

total letters= $75/2.5=30$

6.4 EMBEDDED C CODING

```
#include<reg51.h>

void delay(void)
{
    unsigned int i,j;
    for(i=0;i<3000;i++)
    for(j=0;j<1500;j++)
    {
    }
}

void display(unsigned char c)
{
    switch(c)
    {
        case 'A'://character A
        {
            P0 = 0X7f; delay();
            P0 = 0X88; delay();
            P0 = 0X88; delay();
            P0 = 0X88; delay();
            P0 = 0X7f; delay();
        }break;

        case ' '://space
        {
            P0 = 0X00; delay();
            P0 = 0X00; delay();
        }break;
```

```
case 'T'://character T
{
    P0 = 0X80; delay();
    P0 = 0X80; delay();
    P0 = 0Xff; delay();
    P0 = 0X80; delay();
    P0 = 0X80; delay();
}break;

case 'M'://character M
{
    P0 = 0Xff; delay();

    P0 = 0X20; delay();

    P0 = 0X40; delay();

    P0 = 0X20; delay();

    P0 = 0Xff; delay();
}break;

case 'E' : //character E
{
    P0 = 0Xff; delay();

    P0 = 0X91; delay();

    P0 = 0X91; delay();

    P0 = 0X91; delay();

    P0 = 0X91; delay();
}break
}
void main(void)
{
while(1)
{
display('A');

display(' ');
```

```
display('T');  
display(' ');  
display('M');  
display(' ');  
display('E');  
}  
}
```

Chapter 7

SYSTEM TESTING

7. SYSTEM TESTING

System testing of software or hardware is testing is conducted on a complete, integrated system to evaluate the system compliance with the specified requirements. System testing fall within the scope of black box testing, and as such, should require no knowledge of the inner design of the code or logic.

Hardware Testing

Common test platform, also called an open test standard, is a set of specification defining test methods for diverse components of computer and electronic system to marketed as complete products.

TEST CASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	PASS/FAIL
Test Case 1	Working of 8051 Microcontroller	All ports working properly	All ports working properly	Pass
Test Case 2	Checking RPM of DC Motor	Minimum 500 RPM	Revolution more than 500	Pass

Chapter 8

SNAPSHOT AND RESULTS

8. SNAPSHOT AND RESULT

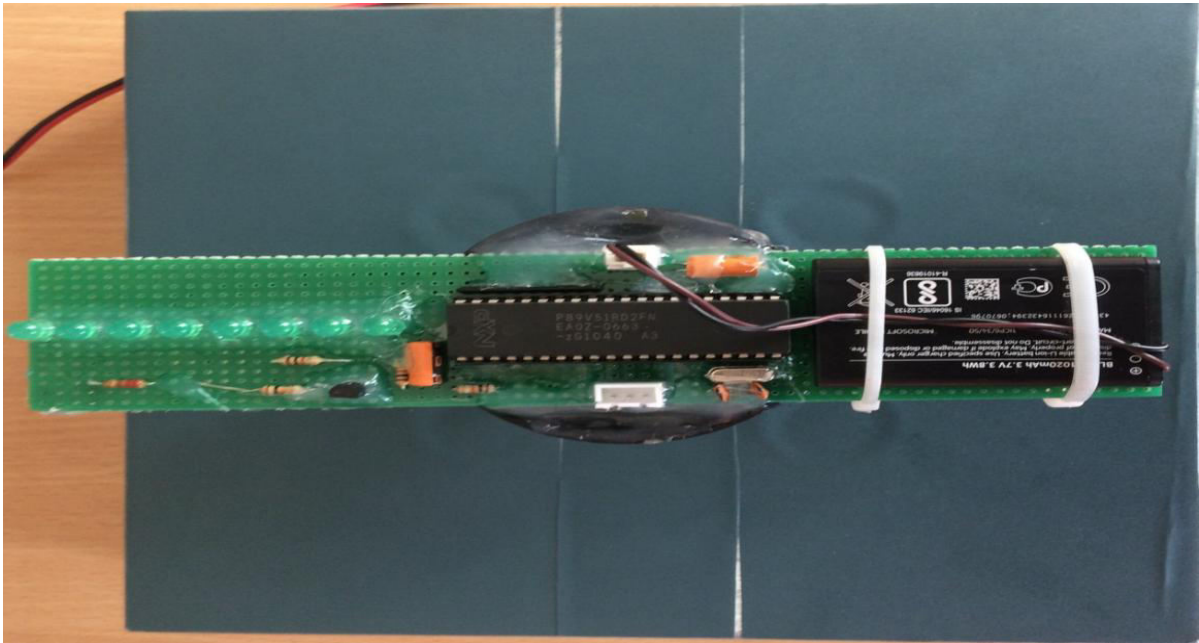


Fig 8.1 Top View of PCB Board

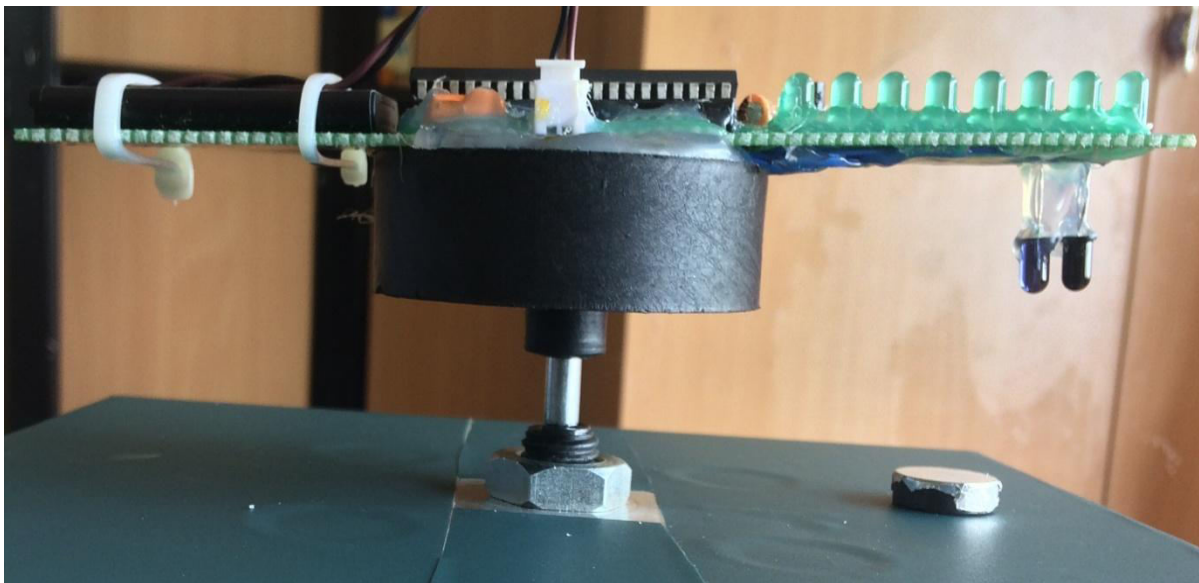


Fig 8.2 Side View of Arrangement

RESULTS

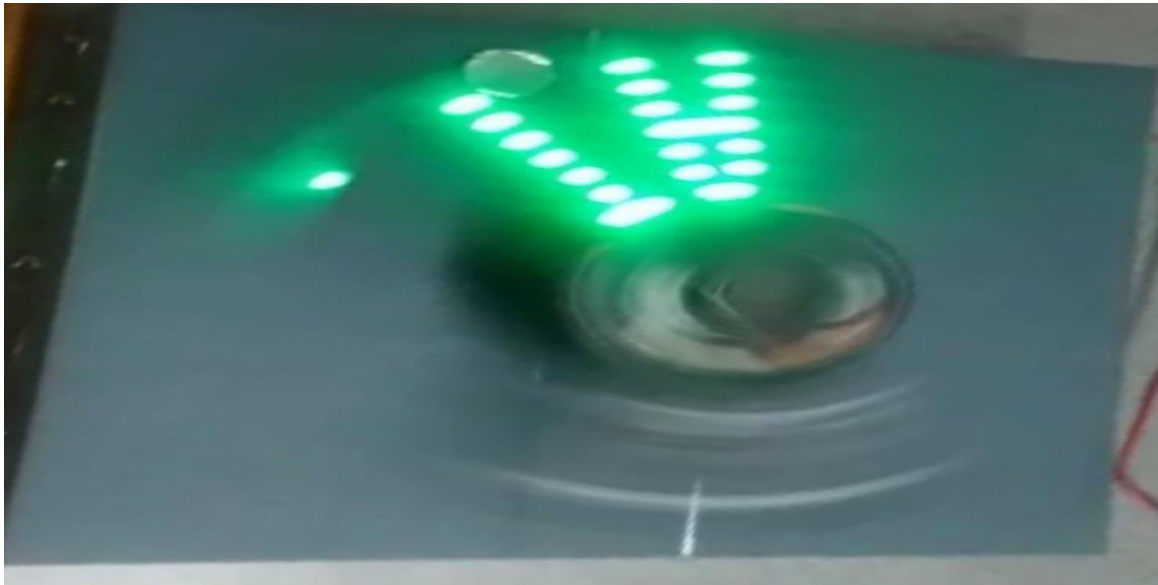


Fig 8.3 Result



Fig 8.4 Result

FUTURE ENHANCEMENTS

FUTURE SCOPE

We can improvise this project in the following ways:-

1. We can make 3-Dimensional version of this display.
2. This display can be programmed as an Analog clock.
3. Real time digital clock can be easily programmed through this.
4. Though we have tried our best to minimize the cost of project but it can further be made cost efficient by cost cutting components.
5. The project can be made more compact with effective arrangement of all the components.
6. Using 3-D version of this display, a very effective representation of earth's globe.
7. We can make two row of LEDs two change the color.

CONCLUSION

CONCLUSION

The proposed system use single columns of LEDs and rotating it in a clockwise so first column pattern must be displayed and some delay must be maintained to control the LEDs ON time of a column. Problem with Existing system is that, we are using each pixel for displaying each pixel individual LED's. This results in huge number of LED even for small display, which in turn increases in cost. Proposed system reduces the number of led which in turn reduce the cost, space multiplexing and crystal clear display with proper assembly of components.

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REFERENCE

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